



Mapping Atmospheric Corrosivity in Shandong

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Abstract Air pollution can significantly accelerate the process of material corrosion, which may cause significant economic losses and serious safety incidents. Atmospheric corrosion maps provide atmospheric corrosivity in a given geographic scope, which can guide the designers to select the most suitable anti-corrosion materials for outdoor projects, also provide useful information for maintenance. This article investigated mapping of atmospheric corrosivity in Shandong Province, China. In order to obtain atmospheric corrosivity data, 100 exposure corrosion test sites were established in Shandong according to International Standard Organization (ISO) 8565. Hot-dip galvanized steel samples were exposed for 1 year in the test sites. Taking the results of exposure corrosion test as the data, inverse distance weighting (IDW) and ordinary kriging (OK) interpolation algorithm were used to estimate the atmospheric corrosivity of Shandong Province according to ISO 9223. The validity of OK and IDW was compared in developing atmospheric corrosion maps of Shandong Province on a 1×1 km resolution. The cross-validation results showed that OK interpolation algorithm with Gaussian semivariogram model get the best result in the prediction of corrosion rate. When the corrosion category was used as the criterion, the IDW interpolation algorithm of power 4 performed best,

predicted results of 74 sites ($n = 100$) were consistent with observed. However, high mean relative errors (MRE more than 37%) and relatively low correlation (R^2 about 24%) indicated that the prediction results of the two interpolation algorithms had a large error, which was caused by the low data density and the complicated corrosive factors of the atmosphere.

Keywords Atmospheric corrosion maps · Interpolation algorithm · Ordinary kriging · Inverse distance weighting · Cross-validation

1 Introduction

Air pollution such as sulfur, nitric oxides, and ozone play important role in the corrosion process observed on various materials (Kambezidis and Kalliampakos 2013). According to statistics, corrosion costs 3–5% of the gross national product for all countries per year, which is greater than the total loss caused by natural disasters and all kinds of accidents (Hou et al. 2017; Koch et al. 2005). Corrosion has also caused numerous public safety accidents (Li et al. 2015). It is generally agreed that 15–35% of corrosion costs can be avoided via appropriate corrosion control practices (Koch 2017). Therefore, it is of great significance to carry out atmospheric corrosivity evaluation. Atmospheric corrosion map is a graphical method to describe the corrosivity of regional atmospheric environment on geographical map, which can help engineers and general users select materials and suitable anti-corrosion coatings. It is the most direct and

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convenient form of regional atmospheric corrosion evaluation results for users. In order to obtain an accurate atmospheric corrosion map, sufficient atmospheric corrosion data is required.

It is well known that ISO 9223 specifies classification, determination, and estimation of atmospheric corrosivity. There are two methods to evaluate atmospheric corrosivity: one is to conduct 1-year exposure test of samples according to ISO 8565, and then measure the corrosion rate of the samples. It is the most accurate and effective method to evaluate atmospheric corrosivity by exposure test. The work on atmospheric corrosion have been reported in many countries for several decades, which include ISO CORRAG (Dean and Reiser 2002; Knotkova et al. 1995), MICAT (Castaño et al. 2010; Morcillo 1995; Pintos et al. 2000; Vilche et al. 1995), ICP/UN ECE Program (Chico et al. 2017; Leuenberger-Minger et al. 2002), etc. These work cost a lot of time and money. The other is to estimate the corrosion rate of standard metals using dose-response functions derived from regression of relevant environmental parameters. This method requires environmental data in the study area, including temperature, relative humidity, chloride deposition, and SO₂ concentration, which play important role in the corrosion process observed on various materials (Cai et al. 2018; Mikhailov et al. 2004; Tidblad et al. 2007).

Due to the easy availability of sufficient data, dose-response functions were used in constructing atmospheric corrosion maps of Switzerland (Leuenberger-Minger et al. 2002; Reiss et al. 2004), Slovakia (Ivaskova et al. 2015), Greece (Kambezidis and Kalliampakos 2013), Istanbul (Karaca 2013), Malaysia (Fathoni et al. 2013), Spain (De la Fuente et al. 2013), and other regions. However, the dose-response functions have a large error as data for atmospheric corrosion maps, most of the input parameters have an uncertainty of approximately 30% (Ivaskova et al. 2015; Leuenberger-Minger et al. 2002). Moreover, in order to obtain more accurate results in different regions, parameters and models of dose-response function have to be adjusted (Mendoza and Corvo 2000; Roberge et al. 2002; Tidblad 2012).

Constructing accurate atmospheric corrosion maps require sufficient and accurate data to support. When the data cannot cover all areas, interpolation is needed in the blank areas (Cole et al. 2012; Huang et al. 2019). In the previous studies, the inverse distance weighting (Cole et al. 2012; Fathoni et al. 2013) and kriging

(Reiss et al. 2004; Kambezidis and Kalliampakos 2013) interpolation algorithms were used in constructing atmospheric corrosion maps, but the validity of the interpolation algorithms were not discussed. In this paper, the atmospheric corrosion maps of Shandong Province were constructed by using the IDW and OK interpolation algorithms based on 100 exposure corrosion test sites; the errors of the two interpolation algorithms were compared.

2 Data and Methods

2.1 Data

In order to carry out the assessment of atmospheric corrosivity, 100 exposure corrosion test sites were set up in Shandong Province, China. Shandong Province is located between latitude 34° 22.9'–38°24.01' N and longitude 114° 47.5'–122°42.3' E, with an area of approximately 1,570,000 km². The peninsula in the east is adjacent to the Pacific Ocean. The specific locations of the exposure corrosion test sites are shown in Fig. 1.

The exposure corrosion test was carried out according to ISO 8565. The samples were 100 mm × 50 mm × 2 mm hot-dip galvanized steel with galvanized layer thickness of about 80 μm. As hot-dip galvanizing technology is the main anti-corrosion technology for outdoor steel structures and widely used in Shandong. In this study, the atmospheric corrosion data were derived from the first-year corrosion rates of galvanized steel samples, which were arranged in 100 exposure corrosion test sites in Shandong Province, and the test period was from June 2018 to June 2019.

Data accuracy is an important factor influencing the estimations of the spatial interpolation methods (Li and Heap 2008). The samples are processed uniformly to ensure the same surface state. Weighing was accurate to 0.1 mg (Mettler Toledo AB135-S). After 1 year of exposure in the test sites, the corrosion products of the exposed samples were removed according to ISO 9226 and re-weighed. The corrosion rate was measured by using the weight loss method, as shown in Eq. (1).

$$r_{\text{corr}} = \frac{\Delta m}{A \cdot \rho \cdot T} \quad (1)$$

where r_{corr} is the corrosion rate, μm/a; Δm is mass loss of the sample, g; A is area of the sample, m²; ρ is the

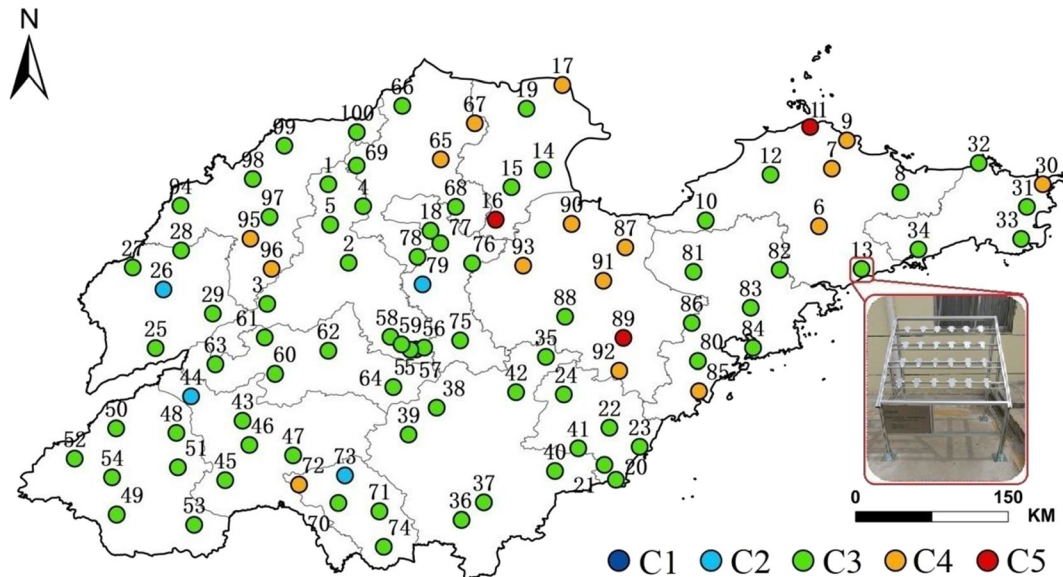


Fig. 1 Distribution and atmospheric corrosivity category of 100 exposure test sites

density of metal, and here means the density of zinc, g/m^3 ; and T is the exposure time, a.

Atmospheric corrosivity was evaluated according to ISO 9223 based on corrosion rate of the standard metallic materials. The corrosion of galvanized steel sample in the first year can be considered as the corrosion of zinc. According to ISO 9223 standard, the corrosivity of atmosphere is divided into 6 categories from low to high: C1, C2, C3, C4, C5, and CX. The corresponding corrosion rate of zinc is shown in Table 1.

Results of atmospheric corrosivity category of 100 exposure test sites in Shandong Province are shown in Fig. 1.

2.2 Methods

In this paper, atmospheric corrosion maps of Shandong Province were constructed by using the IDW and OK interpolation algorithms at a spatial resolution of $1 \times 1 \text{ km}^2$. We compared the performance and validity of two interpolation algorithms. Cross-validation was used to evaluate the results. In order to make the atmospheric corrosion map more accurate and visually better, each category was divided into several sub-categories in constructing corrosion map of Shandong Province. All analysis were accomplished by using the Geostatistical Analyst extension in ArcGIS 10.4.

2.2.1 Inverse Distance Weighting

IDW is one of the simplest and most readily available methods. It is based on the concept of Tobler's first law (the first law of geography), everything is related to everything else, but near things are more related to each other (Chen and Liu 2012; Li and Heap 2008). The IDW is formulated as follows (Eq. (2)):

$$Z_0 = \sum_{i=1}^n Z_i \lambda_i \quad (2)$$

where Z_0 is the estimated value at a predicted point, Z_i is the observed value at point i , and λ_i is the weight value assigned at point i (Eq. (3)).

Table 1 ISO 9223 corrosion rate of zinc for the first year of exposure for the different corrosivity categories

Corrosivity category	Corrosivity	Corrosion rate of zinc r_{corr} ($\mu\text{m/a}$)
C1	Very low	$r_{\text{corr}} \leq 0.1$
C2	Low	$0.1 < r_{\text{corr}} \leq 0.7$
C3	Medium	$0.7 < r_{\text{corr}} \leq 2.1$
C4	High	$2.1 < r_{\text{corr}} \leq 4.2$
C5	Very high	$4.2 < r_{\text{corr}} \leq 8.4$
CX	Extreme	$8.4 < r_{\text{corr}} \leq 25$

$$\lambda_i = \frac{\frac{1}{d_i^p}}{\sum_{i=1}^n \frac{1}{d_i^p}} \quad (3)$$

where d_i is the distance between point i and the predicted point, p is a power parameter, and n represents the number of sampled points used for the estimation.

IDW is concerned with parameters adjustment including the power and search radius. Based on the distribution of the sites, the search radius was fixed at 75 km in this study; too small radius would not be able to construct a complete map. The choice of the IDW power was found to be significant (Kurtzman et al. 2009; Lu and Wong 2008). We evaluated the performance of the IDW of powers 1, 2, 3, and 4 in constructing atmospheric corrosion map of Shandong Province.

2.2.2 Ordinary Kriging

IDW is a deterministic method as compared to geostatistical methods such as kriging (Bailey and Gatrell 1995). On the other hand, kriging is the preferred method in many spatial statistical analyses. OK is the fundamental method in the kriging family. In general, the OK interpolator is based on an assumption of random spatial processes with stationarity, and aims to offer unbiased estimation with a minimized error variance (Cressie 1990; Goovaerts 1997).

OK is similar to IDW in that it is obtained by weighted averaging of neighboring data points. The weights are obtained from the knowledge of the semi-variogram. A variogram expresses the degree of similarity between two observations separated by a given distance. An empirical variogram can be computed from sampled data by using Eq. (4).

$$\gamma(h) = \frac{1}{2n} \sum_{i=1}^n [z(x_i) - z(x_i + h)]^2 \quad (4)$$

where $\gamma(h)$ is the estimated semivariance at a separation distance and $z(x_i)$ and $z(x_i + h)$ are the observed values at x_i and $x_i + h$ separated by h , of which there are n pairs. The basic idea behind a kriging interpolation is to use the variogram to compute weights, which minimize the variance in the estimated value (Wong et al. 2004). Variogram modeling and estimation is extremely important for structural analysis and spatial interpolation (Burrough et al. 2015). There are a number of variogram models that could be employed (spherical, exponential,

Gaussian, liner, etc.), and different variogram models may lead to different interpolations. Thus selecting an appropriate model to capture the features of the data is critical.

In this paper, cross-validation technique was used to choose the best semivariogram model from spherical, exponential, triangular, and Gaussian (Ver Hoef and Cressie 1993). The parameters of these functions were then optimized for the best fit of the experimental semivariogram.

2.2.3 Cross-Validation

Cross-validation was used to evaluate the performance among OK and IDW. The method used here is often referred to as the 'leave one out' method where each site is in turn removed from the interpolation and predicted at the removed site (Denby et al. 2010). The predicted value is then compared with the removed observed value. This is carried out for each station individually and the results can be statistically analyzed.

Several statistical measurements were used to evaluate the performance of two interpolation methods, including of mean error (ME, Eq. (5)), mean absolute error (MAE, Eq. (6)), mean relative error (MRE, Eq. (7)), and root-mean-square error (RMSE, Eq. (8)). The ME measures bias, and the RMSE measures precision (Bekele et al. 2003). If ME, MAE, and MRE are closer to zero, RMSE is smaller, the better the model.

$$ME = \frac{1}{n} \sum_{i=1}^n (Z_i - Z_0) \quad (5)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |Z_i - Z_0| \quad (6)$$

$$MRE = \frac{1}{n} \sum_{i=1}^n \frac{|Z_i - Z_0|}{Z_i} \quad (7)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Z_i - Z_0)^2} \quad (8)$$

where Z_i is the observed value at point i and Z_0 is the corresponding estimated value at the same point.

3 Result

It can be seen from Fig. 1 that among the 100 sites in Shandong Province, the atmospheric corrosion categories of 77 sites are C3, 16 sites are C4, 4 sites are C2, and 3 sites are C5. Based on the first year corrosion rate ($\mu\text{m/a}$) of galvanized steel obtained from 100 exposure test sites, the atmospheric corrosion map of Shandong Province was constructed by using the IDW and OK spatial interpolation method. Cross-validation was used to evaluate the performance.

3.1 IDW

Figure 2 shows the atmospheric corrosion map of Shandong Province constructed by using IDW of powers 1, 2, 3, and 4; the search radius was fixed at 75 km.

From Fig. 2, it can be seen that the atmospheric corrosion categories in Shandong Province are mainly concentrated in C3 and C4, and very few in C2 and C5. The overall distribution of the map is consistent, that is, the corrosivity of the atmosphere in the eastern coastal region (Shandong peninsula) is significantly greater than that in the inland region, which is consistent with the actual observed data. More details were provided by sub-categories and the differences between different regions in the same category were showed. With the change of power, the atmospheric corrosion map in Shandong Province changed significantly in detail. This is mainly due to the relatively high relative humidity and chloride ions in the atmosphere of coastal areas. The increase of relative humidity and chloride ion deposition will significantly enhance the corrosion of the atmosphere, which is consistent with the previous research and ISO 9223 standard (Cole et al. 2011; Mendoza and Corvo 1999).

In order to determine the optimal parameter of IDW, the ME, MAE, MRE, and RMSE of interpolation results at different power values were calculated through cross-validation. The results are shown in Table 2.

It can be seen from Table 2 that the ME is generally positive, indicating that the overall predicted value is slightly smaller than the actual value. In IDW interpolation algorithm, a larger power means the local influence will be accounted for more strongly than influences from more distant locations. ME was the smallest when power was 4. However, MAE, MRE, and RMSE were smallest when power was 1, and increased with the increase of power. So the IDW interpolation algorithm

of power 1 performed best in the prediction of corrosion rate.

Since the corrosion rate of outdoor engineering materials cannot be accurately measured, the atmospheric corrosion category is more concerned by designers in some engineering applications. According to ISO 9223 (Table 1), the interpolation results were compared with the observed atmospheric corrosion category, and the results are shown in Table 3.

It can be seen from Table 3 that, with atmospheric corrosion category as the criterion, the prediction accuracy of IDW was more than 70% in 100 sites. The IDW of power 4 performed best; predictions corrosion categories of 74 sites are consistent with observed. There are only 2 sites that showed a difference of 2 corrosion categories, while others are one category difference in the sites with errors. Therefore, in the case of different criteria, the optimal parameters of IDW are also different.

3.2 OK

Four semivariogram models were compared in constructing atmospheric corrosion maps of Shandong Province, including exponential, spherical, triangular, and Gaussian. ArcGIS 10.4 software was used to optimize the parameters, and the main specific parameters were shown in Table 4.

The atmospheric corrosion maps of Shandong Province constructed by four semivariogram models are shown in Fig. 3. Compared with IDW, the OK interpolation made the atmospheric corrosion categories distribution in the maps more continuous, and there are fewer regions with higher and lower corrosion categories. The overall distribution of atmospheric corrosivity in Shandong Province is consistent with that in Fig. 1; the atmospheric corrosivity in coastal areas is higher than that in inland areas.

The OK interpolation performance of the four semivariogram models was compared by the cross-validation method with corrosion rate data of 100 stations, as shown in Table 5. ME is smallest in the exponential model, indicating the minimum bias. However, the Gaussian model performed best in MAE, MRE, and RMSE. Therefore, the Gaussian model has the highest prediction accuracy in the prediction results of corrosion rate. However, the MRE in the Gaussian model is 37.8297%, which indicates that the interpolation has a large relative error.

Fig. 2 Atmospheric corrosion map of Shandong Province constructed by using IDW of powers **a** 1, **b** 2, **c** 3, and **d** 4

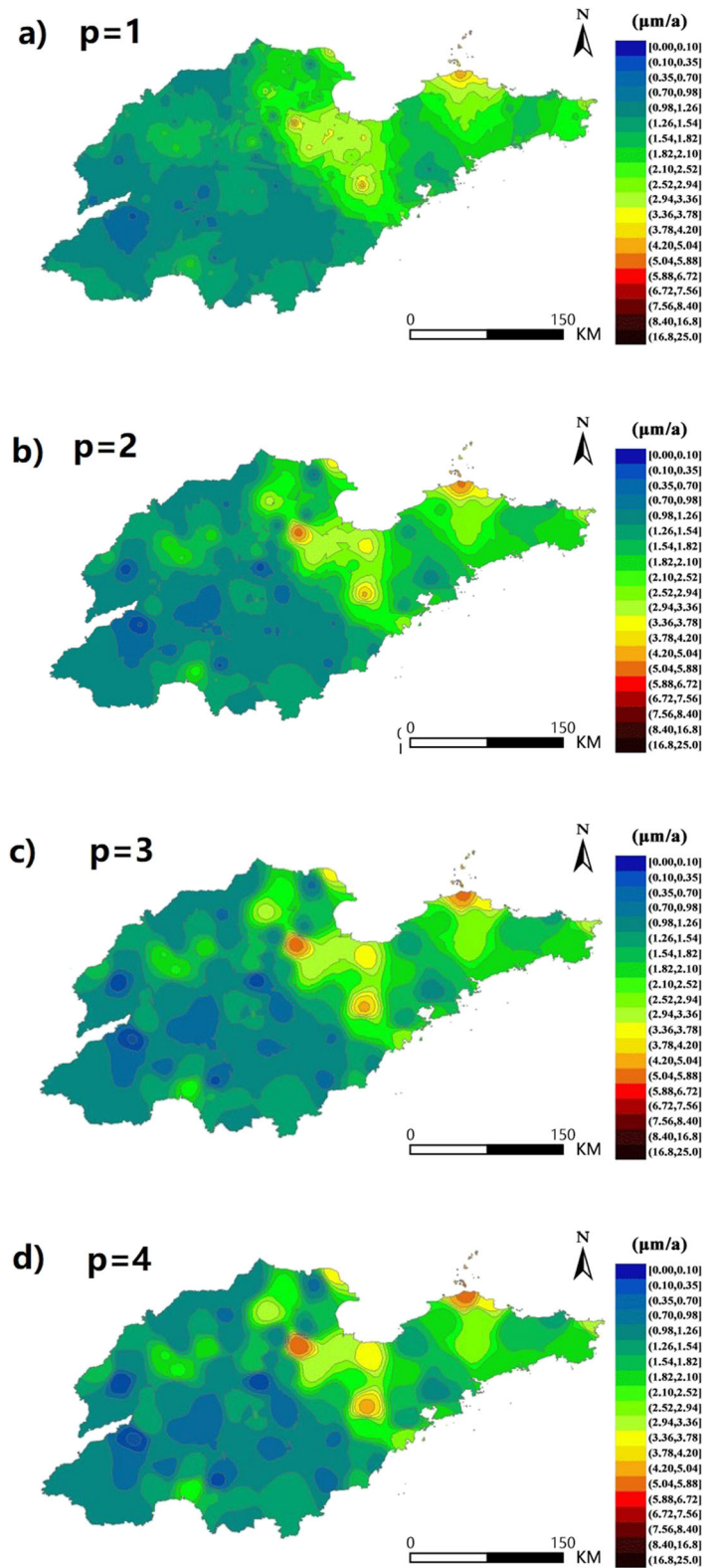


Table 2 Summary of the statistical results for different powers by using cross-validation

Parameters	ME ($\mu\text{m/a}$)	MAE ($\mu\text{m/a}$)	MRE (%)	RMSE ($\mu\text{m/a}$)
$p = 1$	0.0061	0.5597	37.9765	0.8017
$p = 2$	0.0056	0.5845	39.4775	0.8343
$p = 3$	0.0014	0.5986	40.6804	0.8570
$p = 4$	0.0001	0.6138	41.8326	0.8817

Similarly, when the atmospheric corrosion categories were used as criterion, the results are shown in the Table 6. The predictions corrosion categories of 71 sites are consistent with observed in the Gaussian model, and it is the highest of the four models. Therefore, as the corrosion rate and atmospheric corrosion categories were used as the criterion; the Gaussian model had the highest accuracy in constructing atmospheric corrosion maps.

4 Discussion

The distribution of atmospheric corrosion maps of Shandong Province constructed by IDW and OK is same, that is, the atmospheric corrosivity in coastal areas is significantly higher than that in inland areas. According to the results in Table 2 and Table 4, comparing the error of the two interpolation algorithms, the ME, MAE, MRE, and RMSE of OK interpolation with the Gaussian semivariogram model are slightly smaller than IDW ($p = 1$). So, the OK interpolation performs better. When the atmospheric corrosion categories were used as criterion, the IDW of power 4 performs best; there are 74 sites which corrosion categories of predictions results are consistent with observed.

The cross-validation results showed that the sites with the largest errors are No. 16 and No. 89, and there are two categories difference in both interpolation algorithms. From Fig. 1 and Fig. 4, it can be seen that among

the 100 sites, only three sites are C5 (No. 16, No. 89, and No. 11), all of which have large errors, and the errors of No.56 and No. 58 are very small. IDW is based on the first law of geography, and the interpolation results are mainly controlled by surrounding sites. In the OK algorithm, the spatial correlation between sites is described by using semivariance, which depends on the distance between sites. Therefore, the prediction results are also mainly affected by surrounding sites. It can be seen from Fig. 1 that the atmospheric corrosion categories of the surrounding sites of No. 16, No. 89, and No.11 are mainly C3 and C4, with a large difference in corrosion rates. There is an oil refinery near site No. 16, which causes atmospheric corrosivity to be significantly higher than surrounding sites. Site No. 89 is located near the reservoir and the atmospheric relative humidity is high. Compared with the surrounding sites, site No. 11 is closer to the coastline, and the content of chloride ions in the atmosphere is higher. Therefore, a large error was obtained in the prediction process. However, the surrounding sites of No. 56 and No. 58 are all C3, and the corrosion rate is quite close, so the prediction error is very small.

The resulting scatter plots for both IDW and OK interpolation are shown in Fig. 5 and Fig. 6. Compared the cross-validation results of 100 sites, the correlation (R^2) is the largest in the IDW of power 1, which is around 0.2340, and the correlation (R^2) is only 0.1699 in the IDW of power 4. In OK interpolation, the difference of correlation is relatively small, and the correlation (R^2) in Gaussian semivariogram model is the largest,

Table 3 Statistics of sites corrosion categories prediction results by using IDW of different powers

Comparison of corrosion categories	Number of sites			
	$p = 1$	$p = 2$	$p = 3$	$p = 4$
Consistent category	70	73	73	74
One category difference	28	25	25	24
Two categories difference	2	2	2	2

Table 4 The parameters of four semivariogram models of OK

Parameters	Models			
	Exponential	Spherical	Triangular	Gaussian
Nugget	0.3320	0.4738	0.4904	0.5328
Sill	0.4612	0.6342	0.6715	0.2583
Range	1.9359	5.3794	5.3794	1.9460

Fig. 3 Atmospheric corrosion map of Shandong Province constructed by using four semivariogram models of OK, **a** exponential, **b** spherical, **c** triangular, and **d** Gaussian

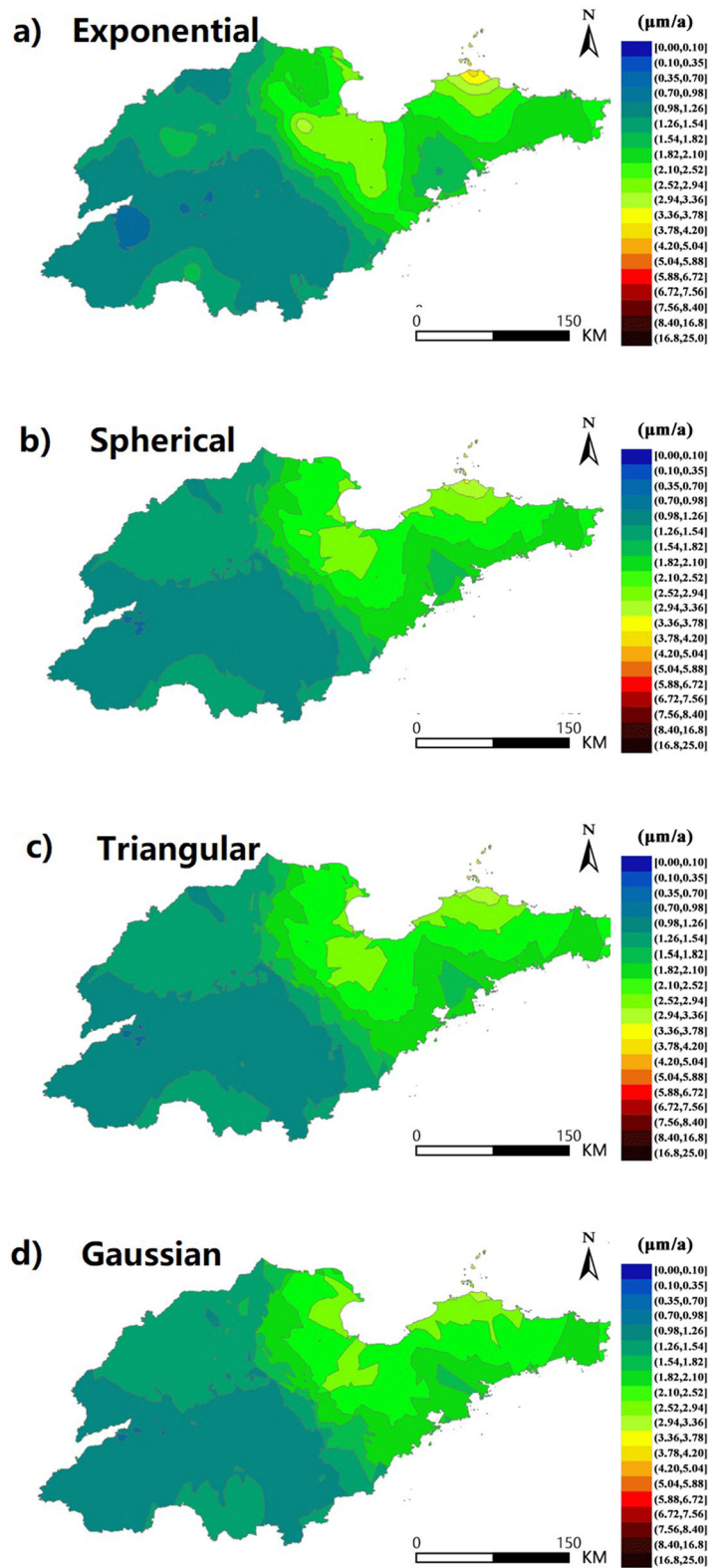


Table 5 Summary of the statistical results for different semivariogram models by using cross-validation

Model	ME ($\mu\text{m/a}$)	MAE ($\mu\text{m/a}$)	MRE (%)	RMSE ($\mu\text{m/a}$)
Exponential	− 0.0032	0.5611	38.0852	0.8039
Spherical	− 0.0065	0.5602	37.8586	0.7995
Triangular	− 0.0060	0.5603	37.8703	0.7996
Gaussian	− 0.0053	0.5597	37.8297	0.7965

which is around 0.2424. However, the correlations of the two interpolation methods are quite low, which is consistent with the results of large MRE calculated by the two methods.

We consider that the errors of the prediction results are mainly caused by two reasons. One is the low data density. The data density may play a role in the performance of the spatial interpolation (Li and Heap 2008). Data variation significantly affects the performance of the spatial interpolation methods. In this study, 100 stations were arranged on the area of 1,570,000 km² in Shandong Province, with an average data density of 6.67×10^{-4} sites per square kilometer. This problem also exists in previous studies. The map of runoff rate for zinc and copper of Switzerland had an average data density greater than 1.52×10^{-3} sites per square kilometer (Reiss et al. 2004). The corrosion map of Korea was constructed with data from 21 sites, with an average data density of 2.0×10^{-4} sites per square kilometer (Kim et al. 2011). Forty sites data were adopted in India corrosion map; the data density was 1.34×10^{-5} sites per square kilometer (Natesan et al. 2005). Atmospheric corrosion map of Chile was constructed by using 1-year exposure data from 31 sites; the data density was 4.1×10^{-5} sites per square kilometer (Vera et al. 2012). Meteorological data of 20 sites were used to establish the dose-response functions, then the atmospheric corrosion map of Athens was constructed by using Kriging interpolation algorithm, with an average data density of 4.85×10^{-2} sites per square kilometer (Kambezidis and

Kalliampakos 2013). None of the above studies carried out the error analysis of interpolation results.

Shandong Province has a large population (about 100 million), and the geographical environment is complex, including coastal areas, mountainous areas, plains, etc. The frequent production activities of human beings and the complicated natural environment lead to large differences in atmospheric corrosivity, which leads to large errors in the constructing of atmospheric corrosion map. For example, the atmospheric corrosivity of site No. 89 is significantly different from the surrounding sites, and the data density is lower, resulting in a larger error in the prediction. In areas with a single natural environment and low human activity, fewer data are needed to construct atmospheric corrosion maps. Increasing data density can significantly improve the accuracy of the corrosion map. The atmospheric corrosivity of site No. 58 is close to that of surrounding sites, the data density is large, and a more accurate prediction result was obtained.

The other reason is that the corrosive factors of atmosphere are complicated. It is well understood now that not only sulfur, but also nitric oxides and ozone, together with temperature, relative humidity, and precipitation, play important role in the corrosion process (Kambezidis and Kalliampakos 2013). The main influence factors of each region are different, but they are not considered in the spatial interpolation algorithms. OK and IDW interpolation algorithms have been successfully applied in the prediction of rainfall and atmospheric environmental

Table 6 Statistics of sites corrosion categories prediction results by using different semivariogram models of OK

Comparison of corrosion categories	Number of sites			
	Exponential	Spherical	Triangular	Gaussian
Consistent category	70	68	68	71
One category difference	28	30	30	27
Two categories difference	2	2	2	2

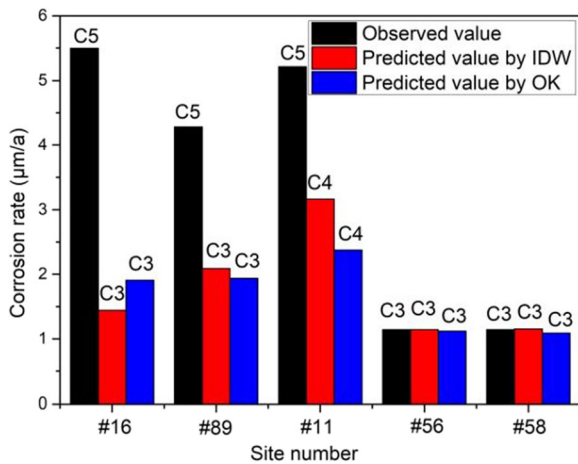


Fig. 4 Prediction results of some sites

pollution (Ahmed et al. 2018; Chen and Liu 2012; Denby et al. 2010; Li and Heap 2014; Wong et al. 2004), but the prediction of atmospheric corrosion is much more complicated.

In this paper, the data obtained by the method of exposing the corrosion sample was used to construct the atmospheric corrosion map of Shandong Province. The accuracy of the data is better than that obtained through meteorological parameters and dose-response functions. Although the cross-validation results showed that the corrosion map had a large error, but only two sites showed more than one category difference in the prediction of atmospheric corrosivity by using both of interpolation algorithms. This is acceptable for the assessment of atmospheric corrosivity, because the corrosion rate of metal materials in

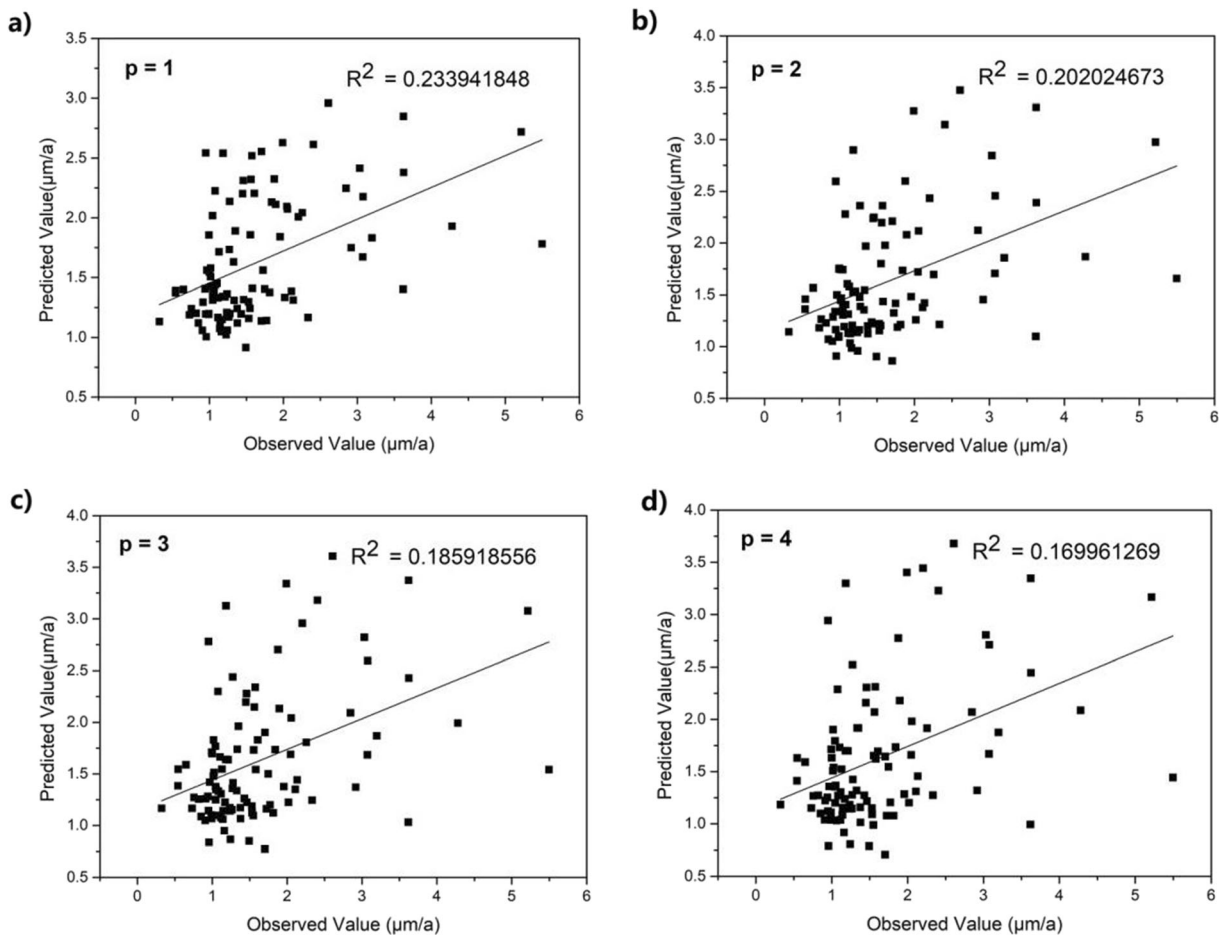


Fig. 5 Scatter plots of IDW interpolation results, **a** $p = 1$, **b** $p = 2$, **c** $p = 3$, and **d** $p = 4$

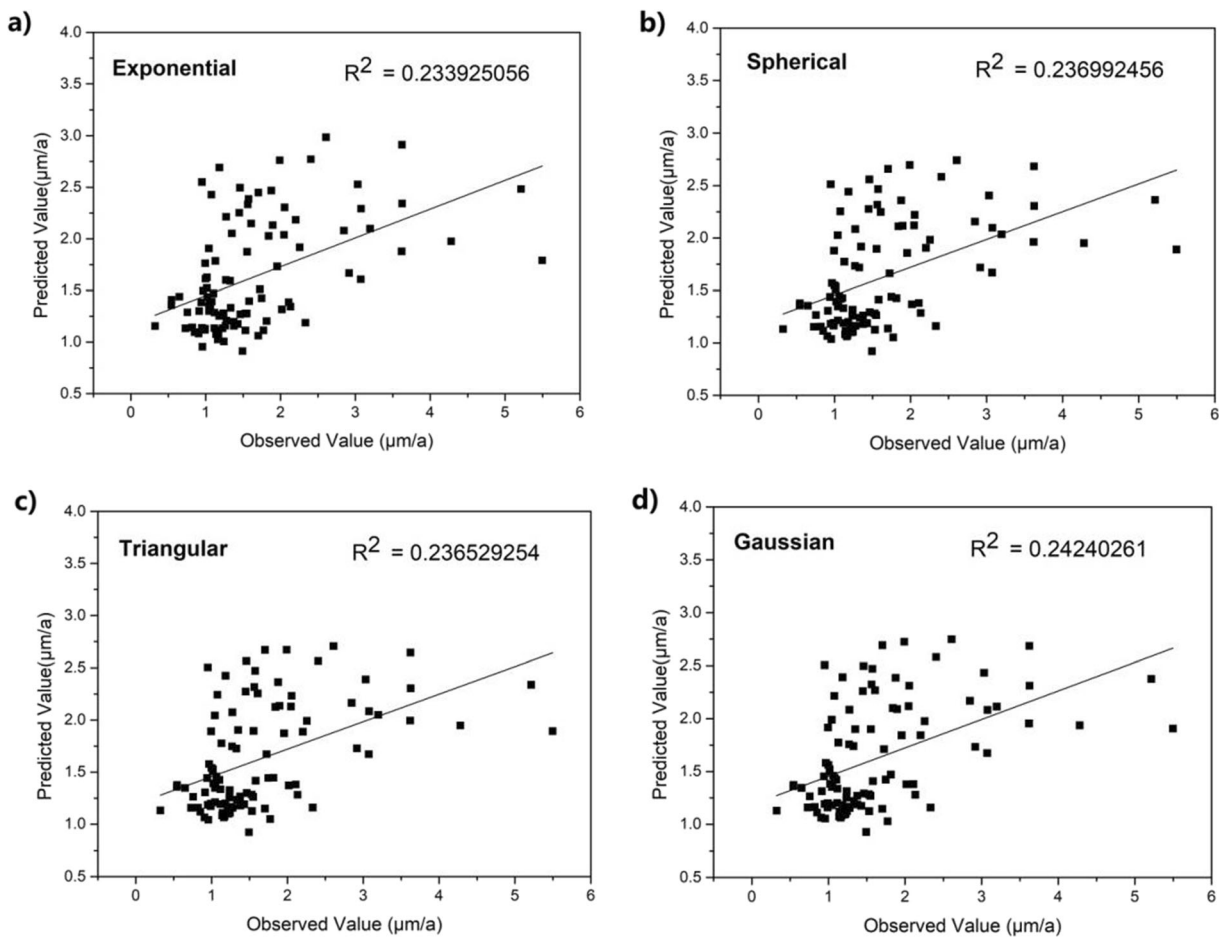


Fig. 6 Scatter plots of OK interpolation results, **a** exponential, **b** spherical, **c** triangular, and **d** Gaussian

atmospheric environment is difficult to accurately predict, and atmospheric corrosion category is more commonly used. The atmospheric corrosion map of Shandong Province constructed in this study can be used to guide the corrosion prevention, operation maintenance, and life prediction for outdoor engineering materials.

5 Conclusions

In this paper, atmospheric corrosion maps of Shandong Province were constructed by using 1 year corrosion rate of hot-dip galvanized steel from 100 exposure corrosion test sites. The errors generated by the IDW and OK interpolation algorithms in mapping were compared. In IDW interpolation, when the power is 1, the error of corrosion rate is the smallest. However, with the atmospheric corrosivity categories specified in ISO

9223 as the criterion, power 4 performed best, predictions corrosion categories of 74 sites are consistent with observed. In the OK interpolation, the prediction result of the Gaussian semivariogram model is the best regardless of corrosion rate or corrosion categories, and the correlation (R^2) is also the highest, which is around 0.2424.

The high error and low correlation are due to the low sampling density points and the complicated atmospheric corrosive factors. It is difficult to accurately apply in constructing atmospheric corrosion map by using the existing interpolation algorithms. As the corrosion rate of metal materials in the atmospheric environment is difficult to accurately predict, and atmospheric corrosion category is more commonly used. The atmospheric corrosion map of Shandong Province constructed in this study can be used to guide the corrosion prevention, operation maintenance, and life prediction for outdoor engineering materials.

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References

- Ahmed, S. O., Mazloun, R., & Abou-Ali, H. (2018). Spatiotemporal interpolation of air pollutants in the Greater Cairo and the Delta, Egypt. *Environmental Research*, 160, 27–34.
- Bailey, T. C., Gatrell, A. C. (1995) *Interactive spatial data analysis*. Vol 413: Longman Scientific & Technical Essex.
- Bekele, A., Downer, R., Wolcott, M., Hudnall, W., & Moore, S. (2003). Comparative evaluation of spatial prediction methods in a field experiment for mapping soil potassium. *Soil Science*, 168, 15–28.
- Burrough, P. A., McDonnell, R., McDonnell, R. A., Lloyd, C. D. (2015) *Principles of geographical information systems*: Oxford university press.
- Cai, Y., Zhao, Y., Ma, X., Zhou, K., & Chen, Y. (2018). Influence of environmental factors on atmospheric corrosion in dynamic environment. *Corrosion Science*, 137, 163–175.
- Castaño, J., Botero, C., Restrepo, A., Agudelo, E., Correa, E., & Echeverría, F. (2010). Atmospheric corrosion of carbon steel in Colombia. *Corrosion Science*, 52, 216–223.
- Chen, F.-W., & Liu, C.-W. (2012). Estimation of the spatial rainfall distribution using inverse distance weighting (IDW) in the middle of Taiwan. *Paddy and Water Environment*, 10, 209–222.
- Chico, B., de la Fuente, D., Díaz, I., Simancas, J., & Morcillo, M. (2017). Annual atmospheric corrosion of carbon steel worldwide. An integration of ISOCORRAG, ICP/UNECE and MICAT databases. *Materials*, 10, 601.
- Cole, I. S., Muster, T., Azmat, N., Venkatraman, M., & Cook, A. (2011). Multiscale modelling of the corrosion of metals under atmospheric corrosion. *Electrochimica Acta*, 56, 1856–1865.
- Cole, I., Corrigan, P., & Hue, N. V. (2012). Steel corrosion map of Vietnam. *Corrosion Science and Technology*, 11, 103–107.
- Cressie, N. (1990). The origins of kriging. *Mathematical Geology*, 22, 239–252.
- De la Fuente, D., Vega, J. M., Viejo, F., Díaz, I., & Morcillo, M. (2013). Mapping air pollution effects on atmospheric degradation of cultural heritage. *Journal of Cultural Heritage*, 14, 138–145.
- Dean, S. W., Reiser, D. B. (2002). *Analysis of long-term atmospheric corrosion results from ISO CORRAG program. Outdoor atmospheric corrosion*. ASTM International.
- Denby, B., Sundvor, I., Cassiani, M., de Smet, P., de Leeuw, F., & Horálek, J. (2010). Spatial mapping of ozone and SO₂ trends in Europe. *Science of the Total Environment*, 408, 4795–4806.
- Fathoni, U., Zakaria, C., Rohayu, C. (2013). Development of corrosion risk map for Peninsular Malaysia using climatic and air pollution data. IOP Conference Series: Earth and Environmental Science. 16. IOP Publishing, pp. 012088.
- Goovaerts, P. (1997). *Geostatistics for natural resources evaluation*: Oxford University press on demand.
- Hou, B., Li, X., Ma, X., Du, C., Zhang, D., Zheng, M. et al (2017). *The cost of corrosion in China*. npj Materials Degradation 1: 4.
- Huang, J., Meng, X., Zheng, Z., & Gao, Y. (2019). Optimization of the atmospheric corrosivity mapping of Guangdong Province. *Materials and Corrosion*, 70, 91–101.
- Ivaskova, M., Kotes, P., & Brodnan, M. (2015). Air pollution as an important factor in construction materials deterioration in Slovak Republic. *Procedia Engineering*, 108, 131–138.
- Kambezidis, H. D., & Kalliampakos, G. (2013). Mapping atmospheric corrosion on modern materials in the Greater Athens area. *Water, Air, & Soil Pollution*, 224, 1463.
- Karaca, F. (2013). Mapping the corrosion impact of air pollution on the historical peninsula of Istanbul. *Journal of Cultural Heritage*, 14, 129–137.
- Kim, Y., Lim, H., Kim, J., Hwang, W., & Park, Y. (2011). Corrosion cost and corrosion map of Korea. *Corrosion Science and Technology*, 10, 3368–3383.
- Knotkova, D., Boschek, P., Kreislova, K. (1995). Results of ISO CORRAG program: Processing of one-year data in respect to corrosivity classification. Atmospheric Corrosion. ASTM International.
- Koch, G. (2017). Cost of corrosion. Trends in oil and gas corrosion research and technologies. Elsevier, pp. 3–30.
- Koch, G. H., Brongers, M. P., Thompson, N. G., Virmani, Y. P., Payer, J. H. (2005). Cost of corrosion in the United States. *Handbook of environmental degradation of materials*. Elsevier, pp. 3–24.
- Kurtzman, D., Navon, S., & Morin, E. (2009). Improving interpolation of daily precipitation for hydrologic modelling: Spatial patterns of preferred interpolators. *Hydrological Processes: An International Journal*, 23, 3281–3291.
- Leuenberger-Minger, A., Buchmann, B., Faller, M., Richner, P., & Zöbeli, M. (2002). Dose–response functions for weathering steel, copper and zinc obtained from a four-year exposure programme in Switzerland. *Corrosion Science*, 44, 675–687.
- Li, J., Heap, A. D. (2008). *A review of spatial interpolation methods for environmental scientists*.
- Li, J., & Heap, A. D. (2014). Spatial interpolation methods applied in the environmental sciences: A review. *Environmental Modelling & Software*, 53, 173–189.
- Li, X., Zhang, D., Liu, Z., Li, Z., Du, C., & Dong, C. (2015). Materials science: Share corrosion data. *Nature*, 527, 441–442.
- Lu, G. Y., & Wong, D. W. (2008). An adaptive inverse-distance weighting spatial interpolation technique. *Computers & Geosciences*, 34, 1044–1055.
- Mendoza, A. R., & Corvo, F. (1999). Outdoor and indoor atmospheric corrosion of carbon steel. *Corrosion Science*, 41, 75–86.
- Mendoza, A. R., & Corvo, F. (2000). Outdoor and indoor atmospheric corrosion of non-ferrous metals. *Corrosion Science*, 42, 1123–1147.
- Mikhailov, A. A., Tidblad, J., & Kucera, V. (2004). The classification system of ISO 9223 standard and the dose–response functions assessing the corrosivity of outdoor atmospheres. *Protection of Metals*, 40, 541–550.

- Morcillo, M. (1995). *Atmospheric corrosion in Ibero-America: The MICAT project. Atmospheric Corrosion*. ASTM International.
- Natesan, M., Venkatachari, G., & Palaniswamy, N. (2005). Corrosivity and durability maps of India. *Corrosion Prevention and Control*, 52, 43–55.
- Pintos, S., Queipo, N. V., de Rincón, O. T., Rincón, A., & Morcillo, M. (2000). Artificial neural network modeling of atmospheric corrosion in the MICAT project. *Corrosion Science*, 42, 35–52.
- Reiss, D., Rihm, B., Thöni, C., & Faller, M. (2004). Mapping stock at risk and release of zinc and copper in Switzerland—Dose response functions for runoff rates derived from corrosion rate data. *Water, Air, and Soil Pollution*, 159, 101–113.
- Roberge, P., Klassen, R., & Haberecht, P. (2002). Atmospheric corrosivity modeling—A review. *Materials & Design*, 23, 321–330.
- Tidblad, J. (2012). Atmospheric corrosion of metals in 2010–2039 and 2070–2099. *Atmospheric Environment*, 55, 1–6.
- Tidblad, J., Kucera, V., Samie, F., Das, S. N., Bhamornsut, C., Peng, L. C., et al. (2007). Exposure Programme on atmospheric corrosion effects of acidifying pollutants in tropical and subtropical climates. *Water, Air, & Soil Pollution: Focus*, 7, 241–247.
- Ver Hoef, J. M., & Cressie, N. (1993). Multivariable spatial prediction. *Mathematical Geology*, 25, 219–240.
- Vera, R., Puentes, M., Araya, R., Rojas, P., & Carvajal, A. (2012). Atmospheric corrosion map of Chile: Results after one year of exposure. *Revista de la Construcción*, 11, 61–72.
- Vilche, J., Varela, F., Acuna, G., Codaro, E., Rosales, B., Fernandez, A., et al. (1995). A survey of Argentinean atmospheric corrosion: I—Aluminium and zinc samples. *Corrosion Science*, 37, 941–961.
- Wong, D. W., Yuan, L., & Perlin, S. A. (2004). Comparison of spatial interpolation methods for the estimation of air quality data. *Journal of Exposure Science & Environmental Epidemiology*, 14, 404.

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